



Supermassive holes

How do black holes get so massive?
Read the answer here:
<http://dgit.in/1a4mQ0g>



First evidence of comet strike

A team of South African scientists discovered the first evidence of a comet entering the atmosphere and exploding 28 million years ago. <http://dgit.in/19jOrGV>

Successful Space Missions to Mars since 1964

Launch Date	Name	Country	Reason
1964	Mariner 4	US (flyby)	Returned 21 images
1969	Mariner 6	US (flyby)	Returned 75 images
1969	Mariner 7	US (flyby)	Returned 126 images
1971	Mars 3 Orbiter/Lander	USSR	Orbiter obtained approximately 8 months of data and lander landed safely, but only 20 seconds of data
1971	Mariner 9	US	Returned 7,329 images
1973	Mars 5	USSR	Returned 60 images; only lasted 9 days
1973	Mars 6 Orbiter/Lander	USSR	Occultation experiment produced data and Lander failure on descent
1975	Viking 1 Orbiter/Lander	US	Located landing site for Lander and first successful landing on Mars
1975	Viking 2 Orbiter/Lander	US	Returned 16,000 images and extensive atmospheric data and soil experiments
1996	Mars Global Surveyor	US	More images than all Mars Missions
1996	Mars Pathfinder	US	Technology experiment lasting 5 times longer than warranty
2001	Mars Odyssey	US	High resolution images of Mars
2003	Mars Express Orbiter/Beagle 2 Lander	ESA	Orbiter imaging Mars in detail and lander lost on arrival
2003	Mars Exploration Rover - Spirit	US	Operating lifetime of more than 15 times original warranty
2003	Mars Exploration Rover - Opportunity	US	Operating lifetime of more than 15 times original warranty
2005	Mars Reconnaissance Orbiter	US	Returned more than 26 terabits of data (more than all other Mars missions combined)
2007	Phoenix Mars Lander	US	Returned more than 25 gigabits of data
2011	Mars Science Laboratory	US	Exploring Mars' habitability

help us understand the planet, its history and hopefully its full potential to help us further our knowledge about what's truly out there.

The National Aeronautics and Space Administration:

NASA has several projects planned over the next seven years to explore Mars. Its present final stage being to set a mechanical rover on the planet's surface in 2020 in pursuit of scientific knowledge. The commissioned projects in the Mars Space Program are:

• MAVEN: (2013-2014)

The Mars Atmospheric and Volatile Evolution (MAVEN) is a reconnaissance mission currently in the works. It involves placing a satellite in Mars' orbit to record data as a way of discovering the history of the planet's atmosphere. One of the features of the MAVEN project is to provide information about the rate at which atmospheric dissipation occurred on Mars. This has several

possible plus-points. The first being that the similarities between our planets may offer a solution to our own environmental issues back here. The second (on a more ideological level) is that we may have a better idea about terraforming Mars to suit our own needs so that we may, at some time in the future, be able to consider it as an option for colonization.

MAVEN is set to launch sometime between 18th November and 7th December, 2013 and be placed in Mars' orbit around 22nd September, 2014.



MAVEN

• InSight: (2016)

The Interior Exploration using Seismic Investigations, Geodesy and Heat Transport. A mouthful of a name, InSight is a project that may just be able to truly explain Mother Nature's existence. The project involves landing a geophysical lander on MARS to study the interior of the planet. Comparing that data with what we already know about our planet, InSight may bring us one step closer to eventually proving (or maybe even disproving) the big bang. InSight is programmed to dig its sensors beneath the surface of Mars, detecting the remnants of the processes of terrestrial planet formation, as well as measuring what NASA describes as the planet's "vital signs": Its "pulse" (seismology), "temperature" (heat flow probe), and "reflexes" (precision tracking).

InSight is scheduled to launch sometime in March 2016 and land on the planet's surface in the September of the same year.

• Mars 2020:

The Mars Rover program set for 2020 is one in a long line of attempts to successfully perform a mission of robotic exploration of Mars. Not much is confirmed about this mission since its budget depends on the future success of NASA. To keep mission costs low and decrease the risks this highly-capable rover would ideally be based on NASA's successful Mars Science Laboratory design, including the guided entry and sky-crane landing system that successfully proved its worth when it carried the Curiosity rover to the planet's surface back in August, 2012.

The European Space Agency and The ExoMars Program: (2016-2018)

The ESA is a joint effort of twenty states in the European Union. It was formed in 1975, by effectively merging two older organizations: ESRO (European Space Research Organization) and ELDO (European Launcher Development Organisation). Since then, they have had their share of highs and lows as space explorers. However, their ExoMars Program has a high possibility of giving them (and through them, us all) a huge push in the right direction.